USE-CASE FLOW SPECIFICATION
Food Bank Surplus Redistribution Platform

Name: Reia S | SRN: PES2UG24CS921

Use Case: Reserve Food Batch and Pickup Window (UC-03)

Primary actor: Shelter Coordinator

Supporting actors: Donor Partner; Notification Service

Preconditions

1. The shelter coordinator is registered, verified, and signed in. 2. At least one suitable food batch is active and has not expired. 3. The batch has not already been reserved.

Postconditions

Success: The batch is marked Reserved, the pickup window is recorded, a unique pickup verification code is generated, and the shelter and donor receive confirmation. **Failure:** No reservation is created; the batch remains available if it has not expired.

Main Success Scenario

1. The shelter coordinator opens the available food-batch board.

2. The system displays active batches with quantity, expiry countdown, location, and dietary tags.

3. The coordinator searches or filters batches and selects a suitable batch.

4. The system displays batch details and available pickup windows.

5. The coordinator selects a pickup window and submits the reservation.

6. The system checks that the coordinator is verified and the batch is available and unexpired.

7. The system reserves the batch and records the pickup window.

8. The system generates a unique pickup verification code.

9. The system notifies the shelter coordinator and donor partner.

10. The system displays confirmation; the use case ends successfully.

Alternate Flow — Batch No Longer Available

6a. If the batch expires or another shelter reserves it before step 7, the system rejects the reservation and does not create a duplicate claim.

6a1. The system displays: “This food batch is no longer available.”

6a2. The system refreshes the active batch list and allows the coordinator to select another batch. If no suitable batch remains, the coordinator exits without a reservation.

Related requirements: FR-002, FR-003, FR-004, FR-005, NFR-002.